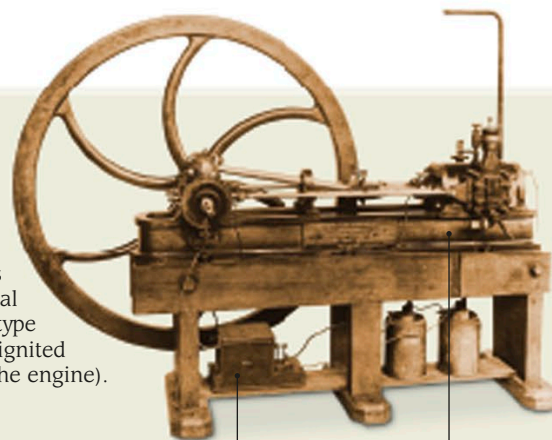


The story of engines

An engine is a machine with moving parts that converts one form of energy, normally chemical energy from burning fuel, into the energy of motion. The invention of the steam engine, which harnessed the power of steam to operate machinery in mills and factories, gave rise to the Industrial Revolution. By the early 1800s, steam engines were also driving ships and locomotives. The invention of the gasoline-fueled internal combustion engine and the jet plane engine revolutionized transportation still further.

Gas engine

In 1860, Belgian engineer Etienne Lenoir designed an engine that generated power by burning gas and air inside a cylinder. It was the first successful internal combustion engine (the type of engine where fuel is ignited inside the engine).



Electric-spark ignition system

Gas and air mix inside the cylinder.

Exhaust gases escape through smokestack.

Flywheel is turned by power from the piston, which drives the crank.

Crank is linked to crosshead by a connecting rod.

Cogwheels are turned by crank, creating forward motion.

Boiler heats water to create steam that moves piston.

Piston rod connects the piston to the crosshead.

Crosshead is linked to flywheel by a connecting rod.

First steam locomotive

British engineer Richard Trevithick's *Penydarren* locomotive made the first successful train trip of 9 miles (14 km) at an ironworks in South Wales, UK, on February 21, 1804. The boiler sprang a leak on the return trip, and the engine was removed from the locomotive. The engine was later used as a stationary engine to drive a heavy hammer at the same ironworks. This model is based on Trevithick's drawings.

Model of Trevithick's *Penydarren* locomotive

Key events

c 50 CE

Greek inventor Hero of Alexandria designed a device that used steam to spin a sphere. It had no practical purpose.



Hero's device

1712

British inventor Thomas Newcomen built the first practical steam engine for commercial use. It used low-pressure steam inside a large cylinder.

1765

James Watt designed a much more efficient steam engine, which also converted the linear (up-and-down) movement of the pistons into a rotary motion.

1804

Richard Trevithick mounted a compact, high-pressure steam engine on wheels to create the world's first steam locomotive.